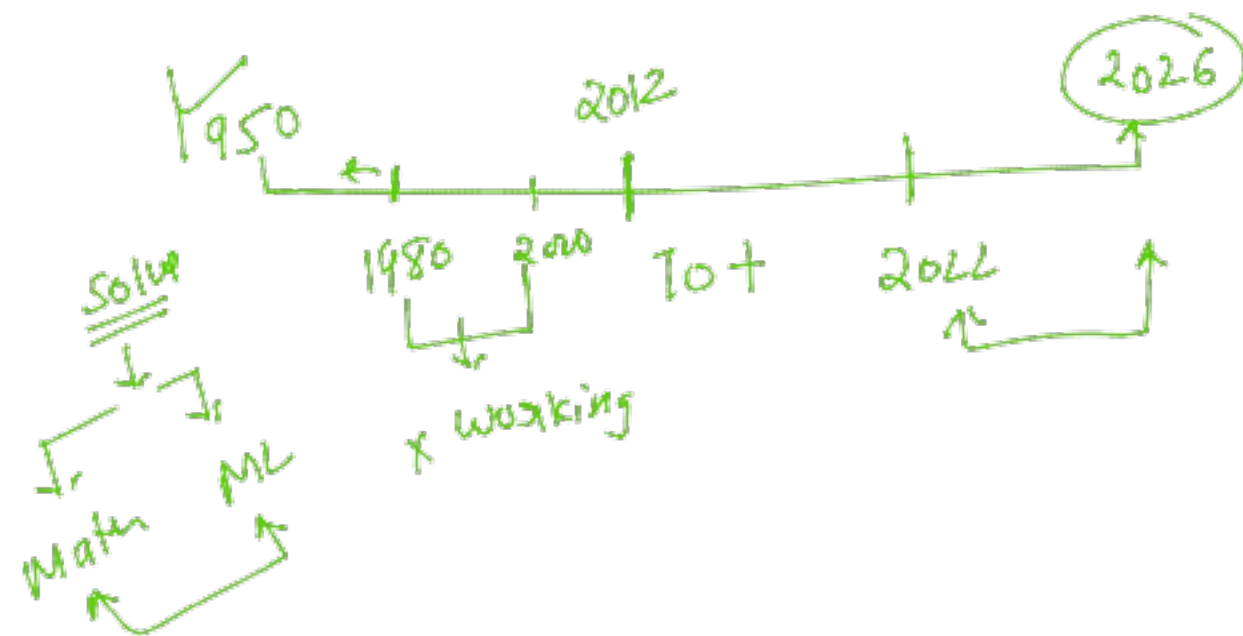




Problem Statement
⇓

'1990'

↑
Data Sci → Bank
↑
II.

python

☺
↑
=

Information

C1 → [20, 750, Mumbai, 28]

⇓

Approve ☒

Reject ☐

→ C2 → [9, 550, Pune, 34]

⇓

Approve ☐

Reject ☐

in a day

5 Request

↓

5 customer's data

↓

manually → You can check

↓

✓
x

But → in a day

[1000 Request

1000 custom

Single { 10 days }



CS → [9, 550, M, 34]

→ 8 day

1000

↓

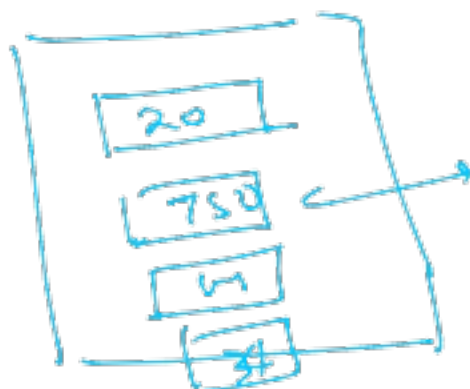
10,000 / m

↓

Fullfilling

↓

HDFC



Stater

↓

✓

x

they thought → let Build Rule

	income	c-s	city	Age	status
C1	—	—	—	—	Yes
C2	—	—	—	—	No
C3	—	—	—	—	Yes
C4	—	—	—	—	No
C5	—	—	—	—	—
⋮	⋮	⋮	⋮	⋮	⋮
Goal	—	—	—	—	↑



Rule
↓
if cibil-score > 750
and income > 30
→ Yes

else
→ Reject

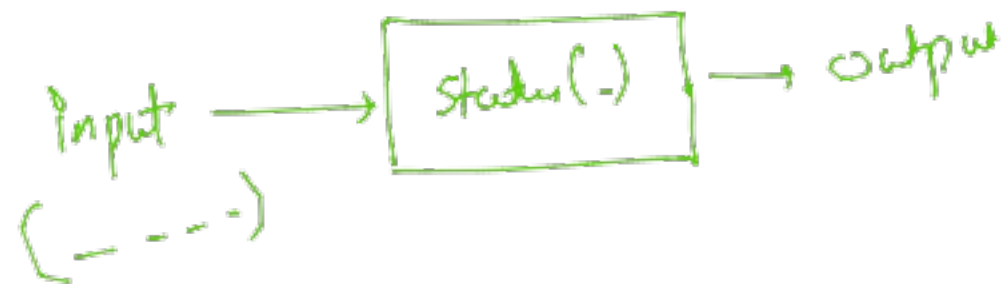
python
fn

```

def status(cibil-score, income):
    if (cibil-score > 750) and income > 30:
        if city:
            return Approved
    else:
        return Reject

```

T-F ✓
 T-II
 T-III



eg.

CI

↓

civil-sion = 749 (-1)] x x

score = 70
v.v. good

→ com som idea

↓

Big

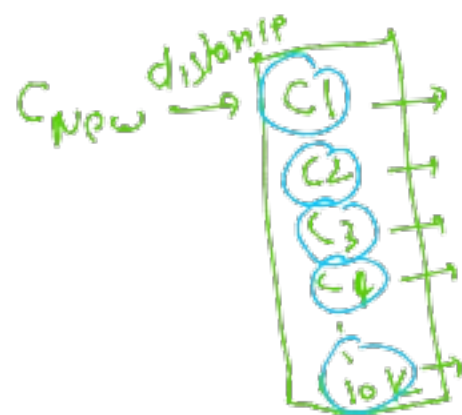
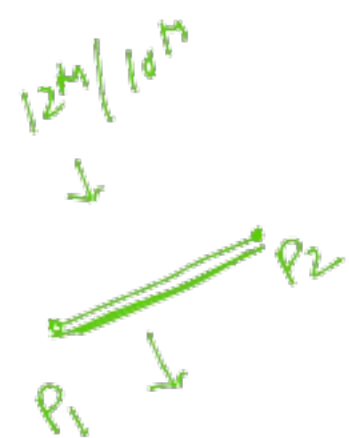
Quar

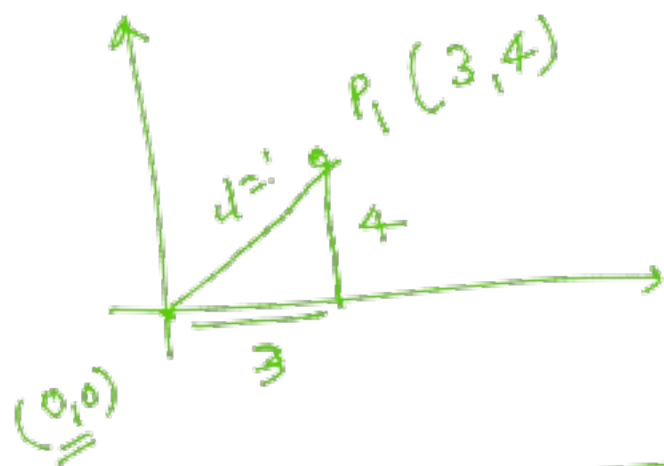
MATH ←

Red

→

→

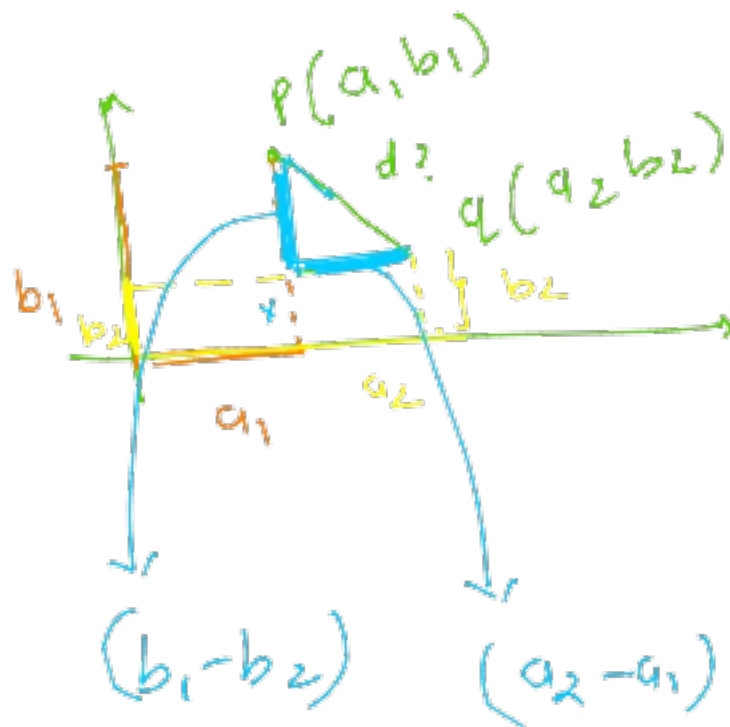




$$d = \sqrt{9 + 16}$$

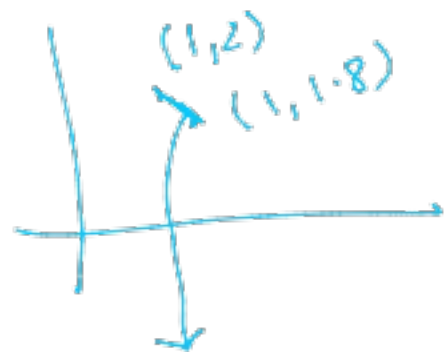
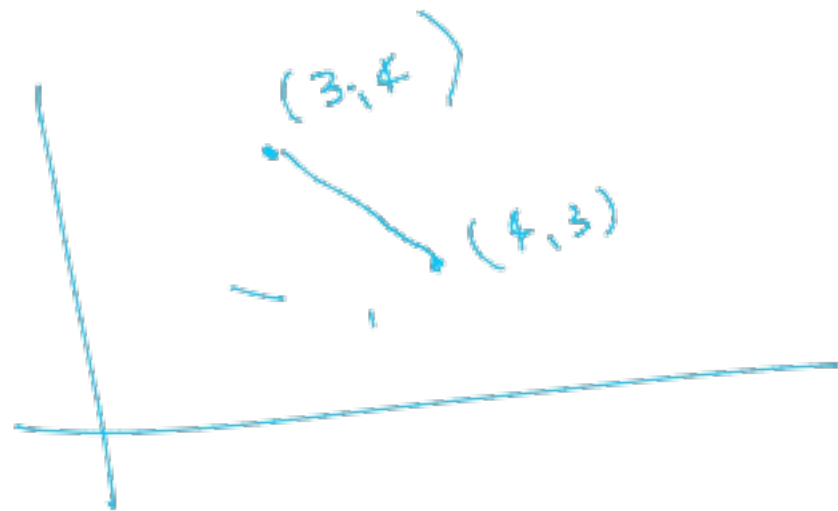
$$d = \sqrt{25}$$

$$\boxed{d = 5}$$



$$d = \sqrt{(a_2 - a_1)^2 + (b_1 - b_2)^2}$$

$$\rightarrow d = \sqrt{(a_1 - a_2)^2 + (b_1 - b_2)^2}$$



distance ↓

$$d = \sqrt{(-1)^2 + (-1)^2}$$

$$d = \sqrt{2}$$

$$d = 1.41$$

Ans

$$d = \sqrt{(a_2 - a_1)^2 + (b_2 - b_1)^2}$$

	c1	c2	c3	c4
r1	3	4	5	4
r2	4	5	6	4

↓
p(a₁ a₂ a₃)

q(b₁ b₂ b₃)

p(3, 4, 5)

q(4, 5, 6)

↓ distance

$$d = \sqrt{(-1)^2 + (-1)^2 + (-1)^2}$$

$$d = \sqrt{3}$$

Linear Alge¹
11

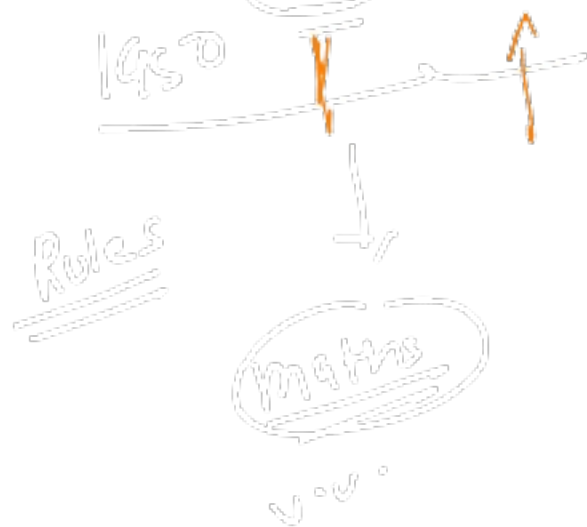
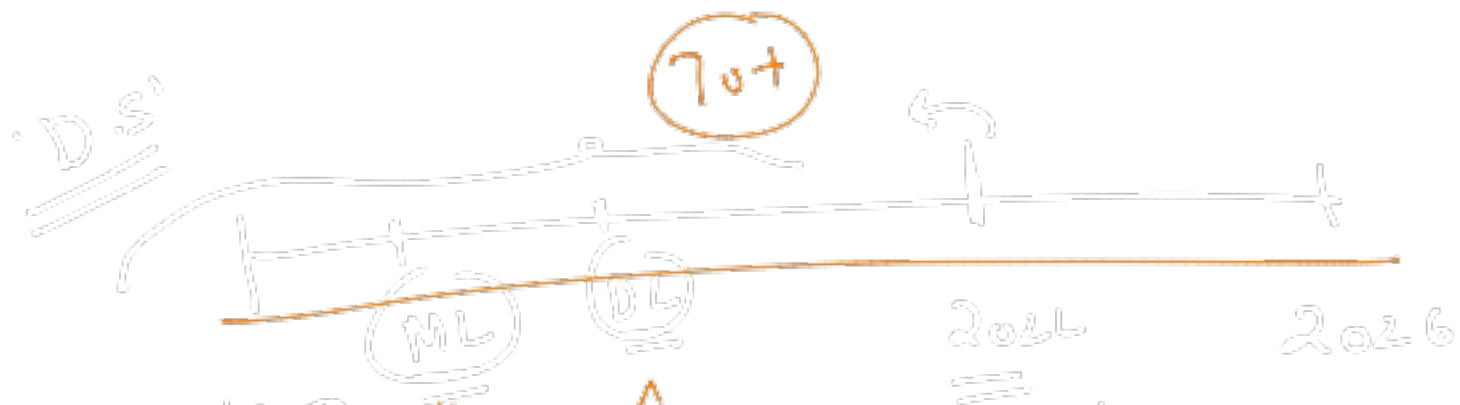
$$p = (a_1 \ a_2 \ a_3 \ a_4 \ \dots \ a_n)$$

$$q = (b_1 \ b_2 \ b_3 \ b_4 \ \dots \ b_n)$$

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2}$$

$$d = \text{sqrt} \left(\sum_{i=1}^n (a_i - b_i)^2 \right)$$

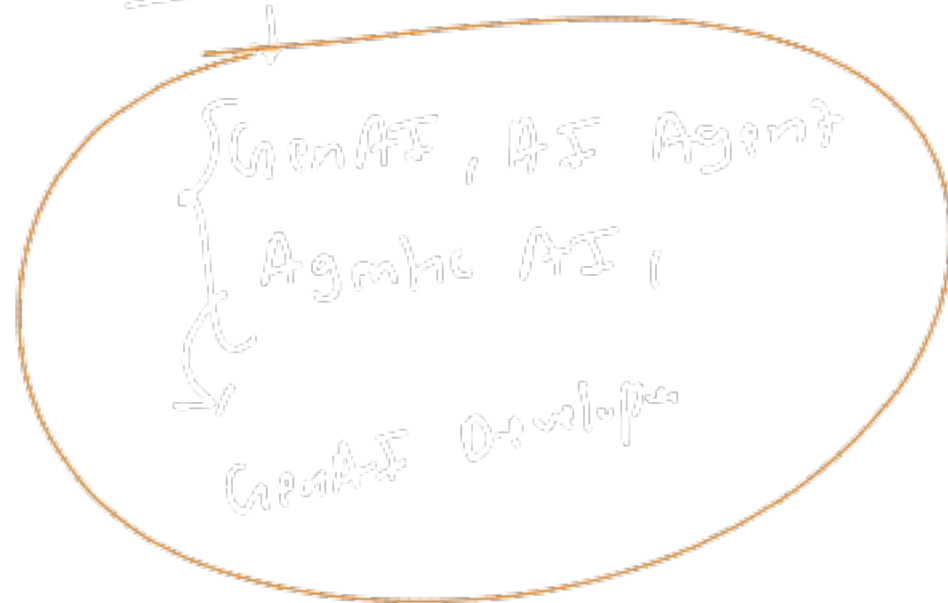
↑
python



.net

'python'

↑



Solve

101c custom 's data

U,

Build

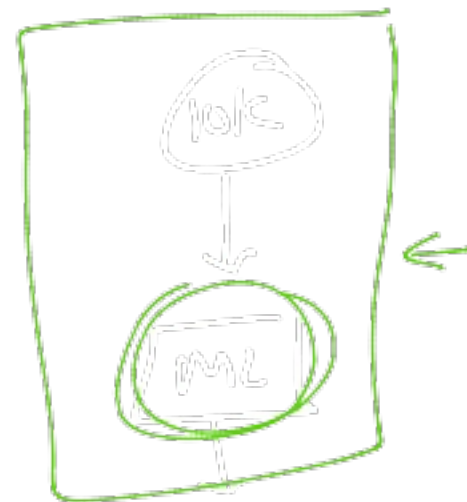
Good

Bad

Rule

U,

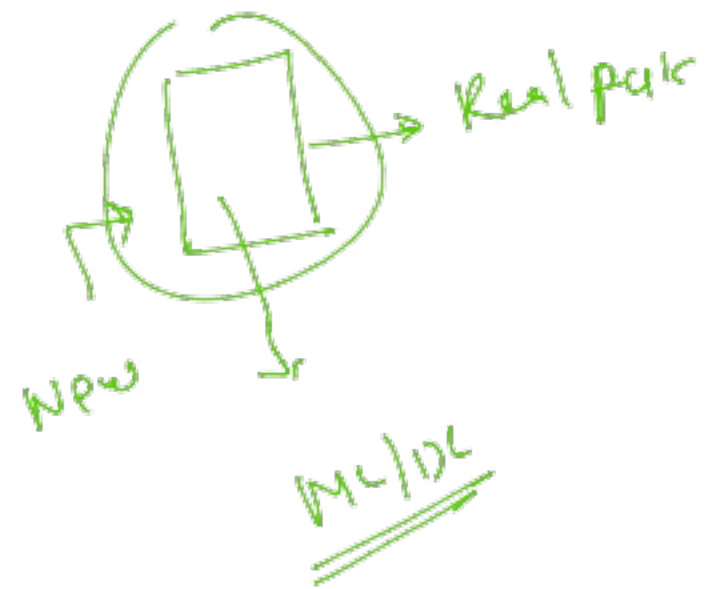
ML



M() ← future

↓
good/B

10k images



20k



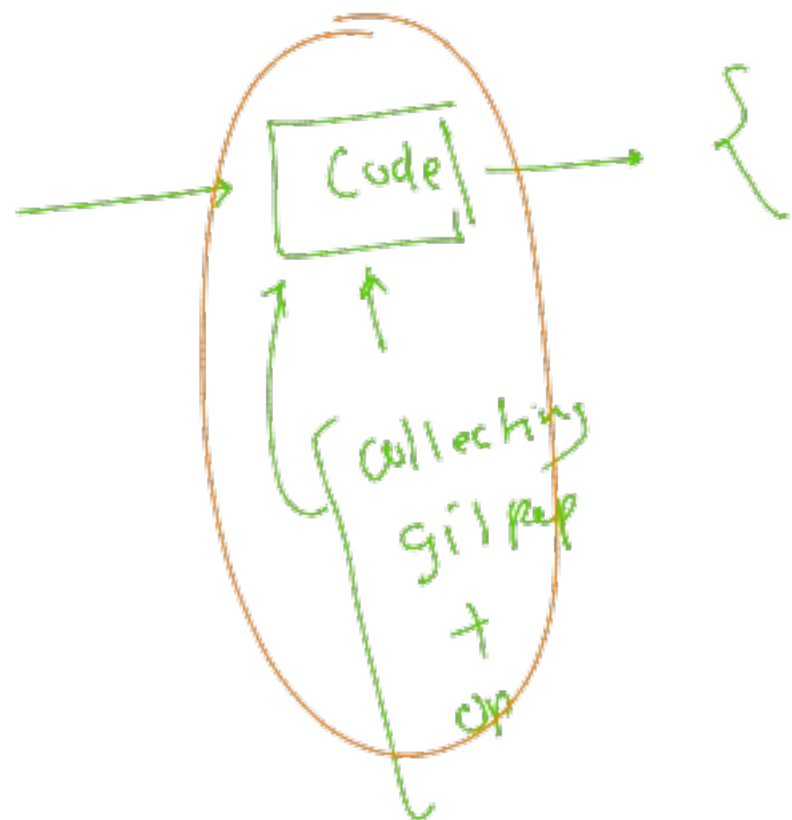
MC

'train' ← [image
quality]

'pretrained' →

Rahul

↓
Collect te int



python

math
(10th)

sklearn

pytorch

ML

DL

Chd GPT
11.

GenAI
(2022)

2017

Google

map

NLP
(text)

Vision

Comp

(chatGPT tools)
solutions
P.E
RAH

2023

python

js

Lang chain

2024

Openai



consume paper for gen

collect
(w, ...)



trained



GPT-1.0

Open



θ

GPT-2.0 → Open

A



A

θ



GPT-3.0



A

ChatGPT

2022

2018



2019/2020

Size ↑

Timeline
2025

Costs

3TB

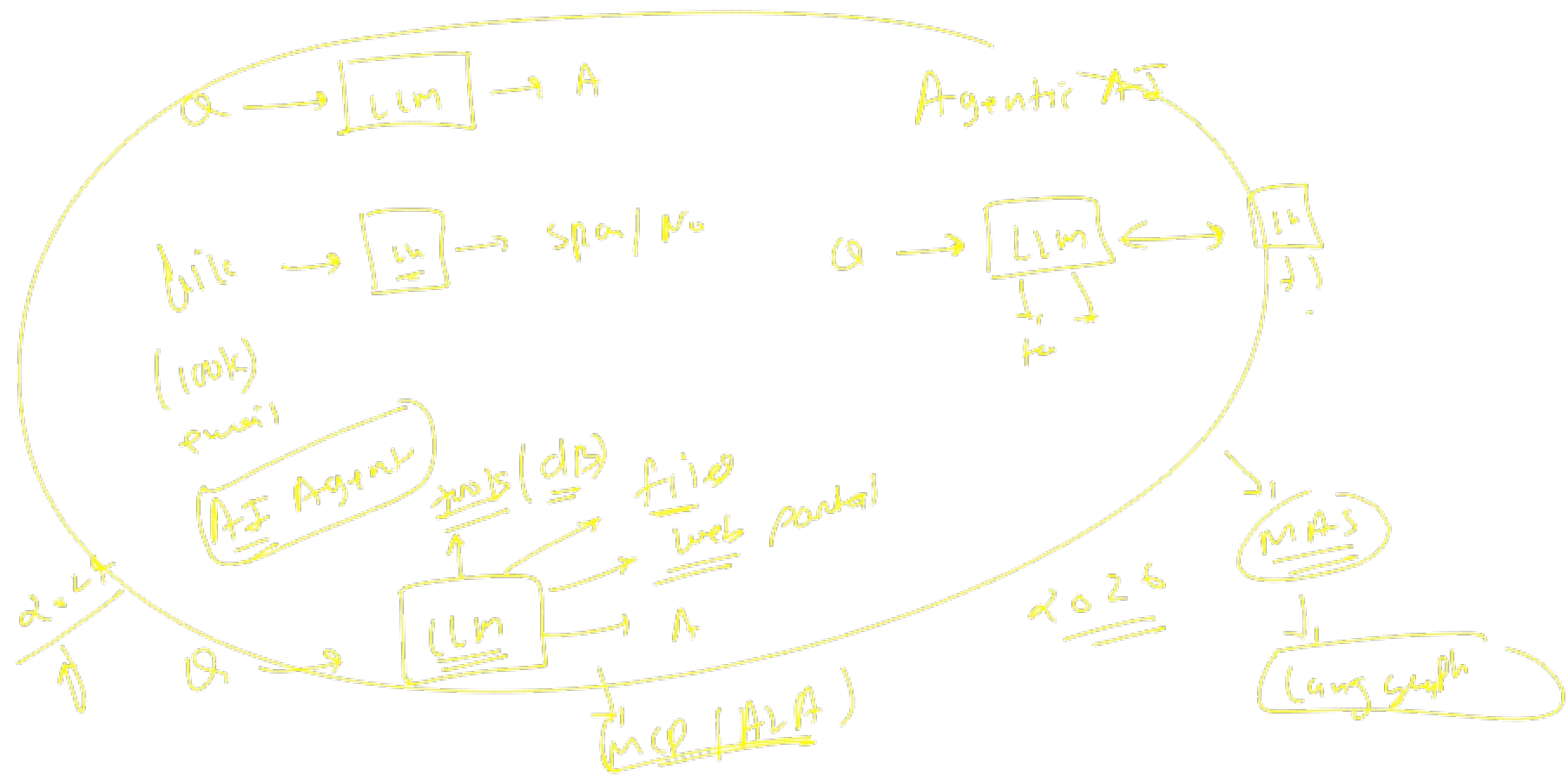
train

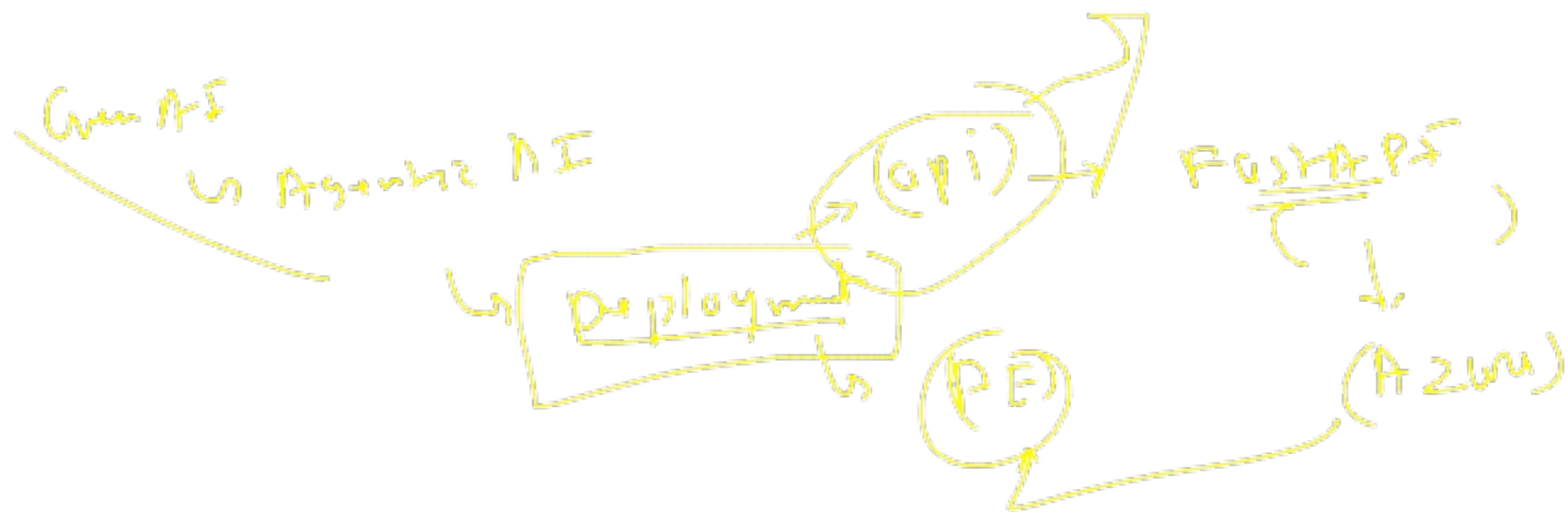
CPU
1/5 hr

200-300
year

Openai
next
year

train (1016 GPU)





→ Full stack →

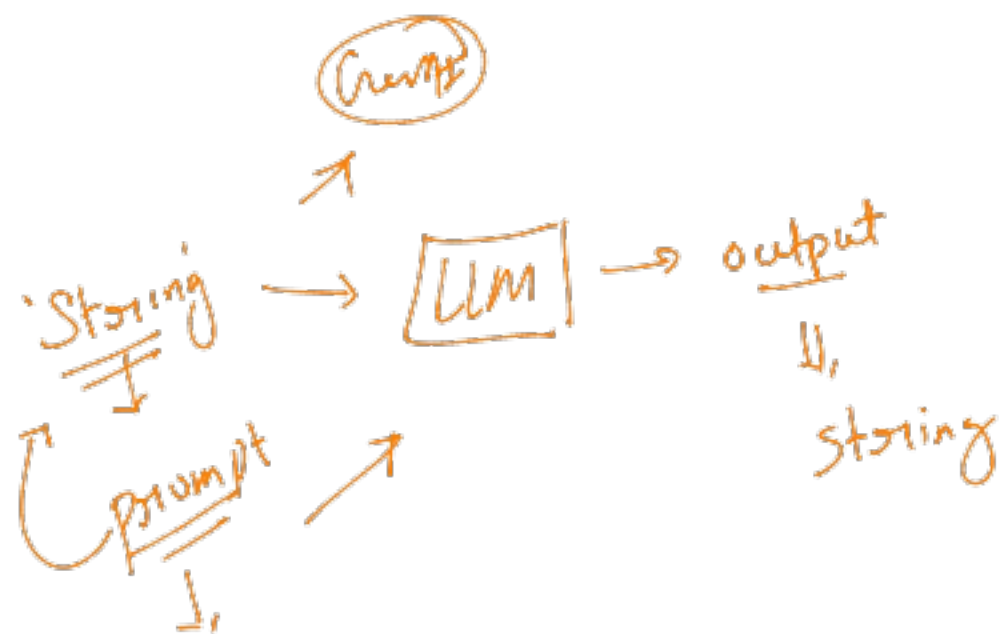


ChalBot
+
memory

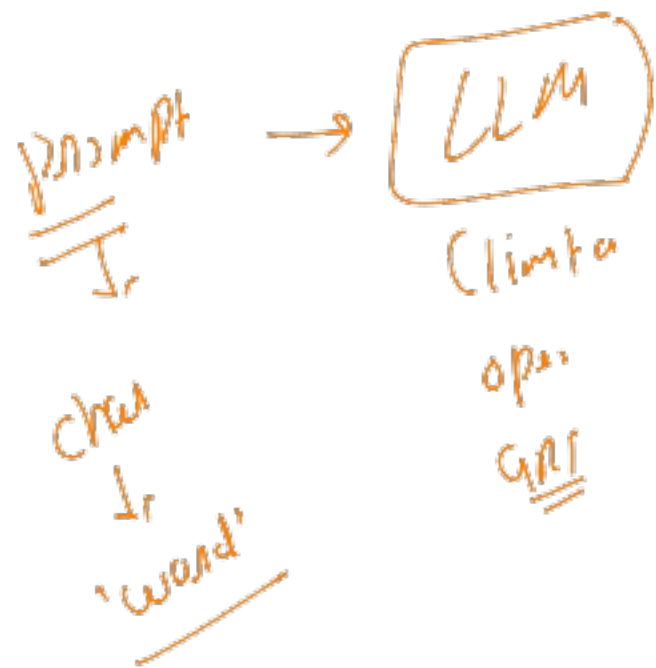
+
tool integration

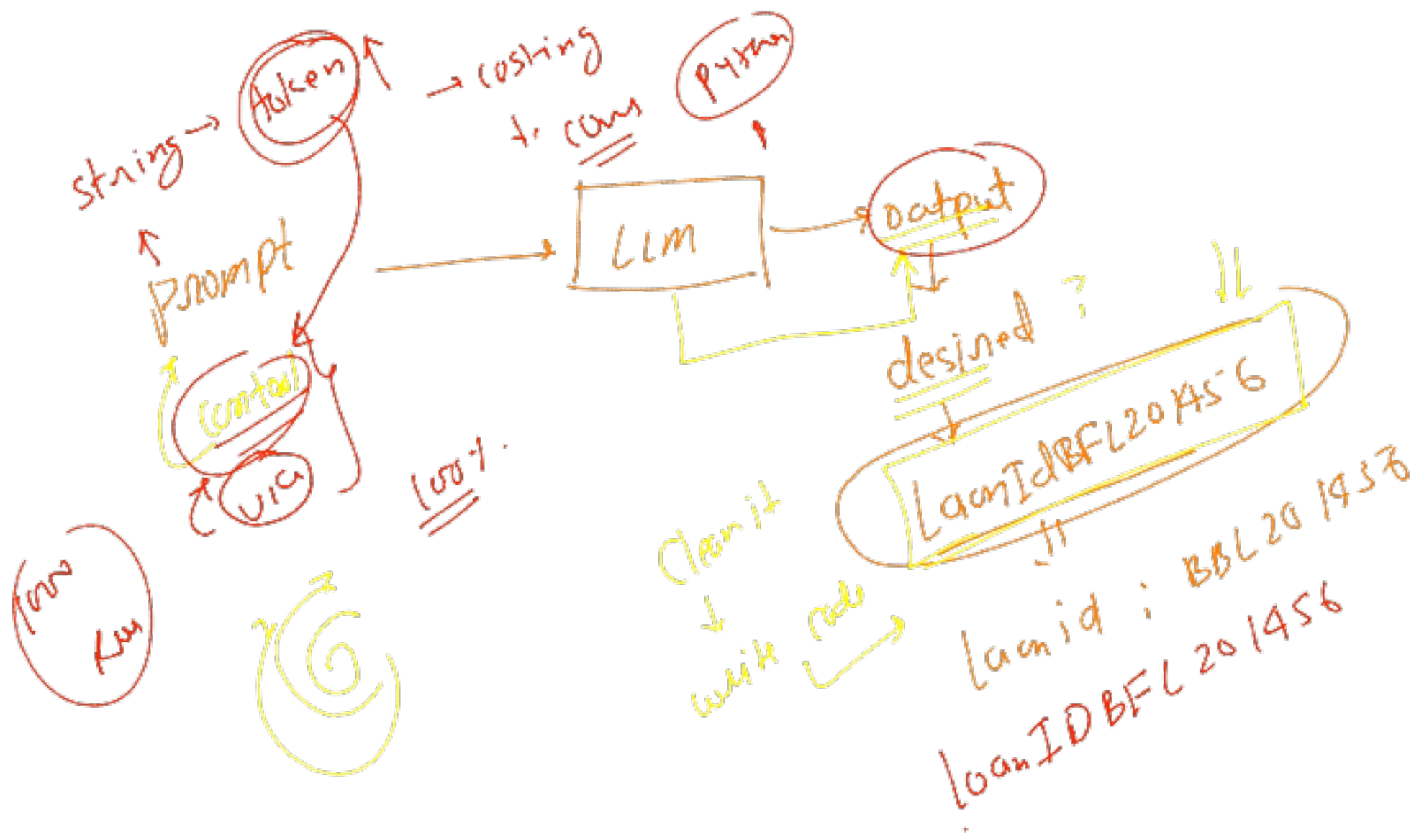
||

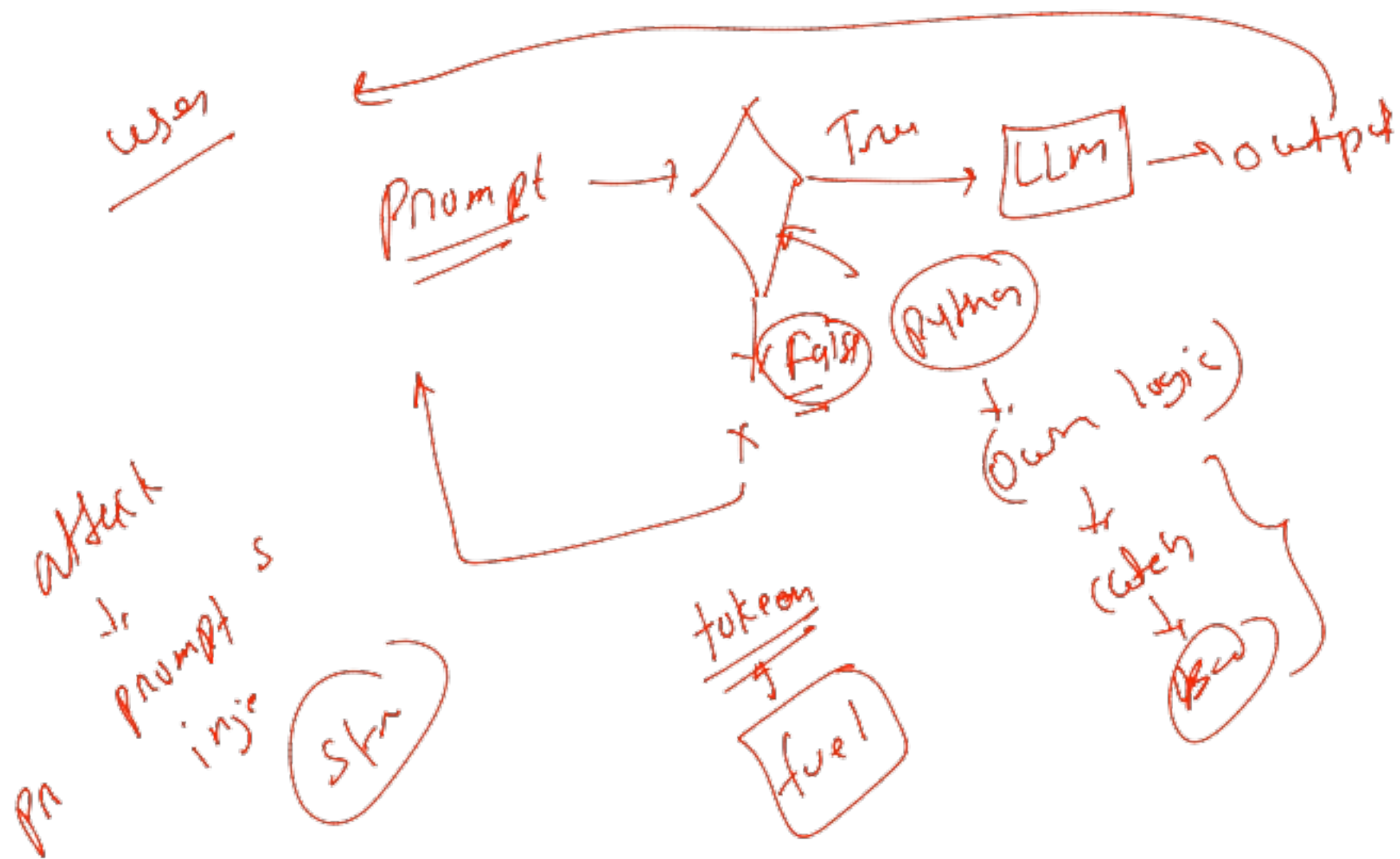
External (python functions)
W.
dB M2
etc

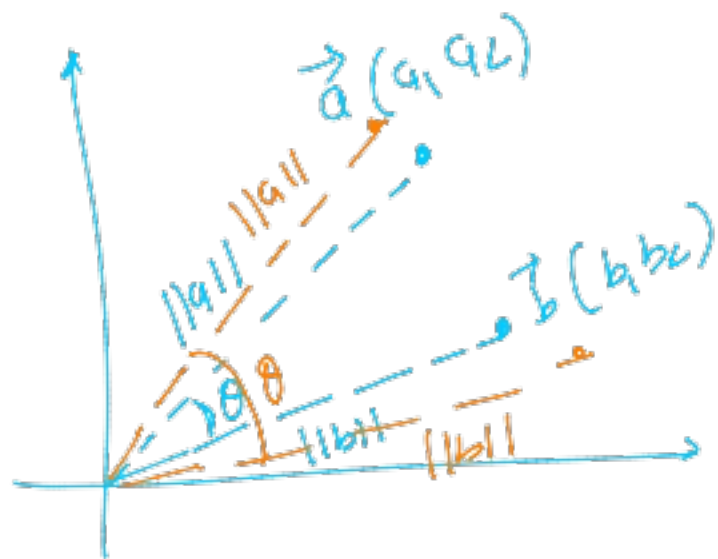


String
↓
method →

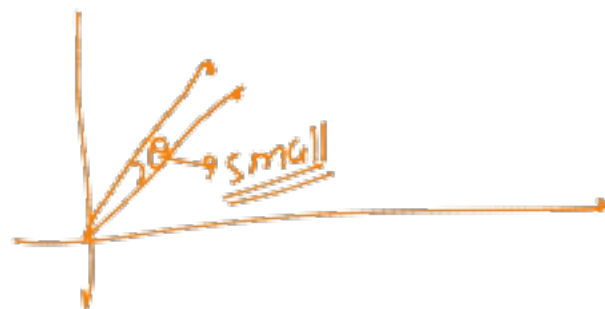








(case ii)

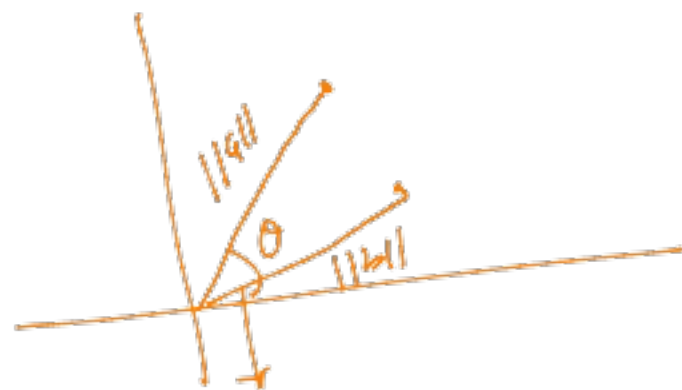


(case -i)

points are ~~get~~ going away from each other

↓
 $\theta \uparrow$
=

test match / test
 $\uparrow$
 co-sine-similarity
 $\downarrow$



$\theta = 0$

$\theta \uparrow$ not similar

$\theta \propto \frac{1}{\text{similarity}}$



$(\theta \uparrow \text{ similar } \downarrow)$

$\theta \downarrow \text{ sim } \uparrow$

$$a = [a_1 \ a_2 \ a_3 \ a_4 \ \dots \ a_n]$$

$$b = [b_1 \ b_2 \ b_3 \ b_4 \ \dots \ b_n]$$

- (i) $\vec{a} + \vec{b} \Rightarrow$ component wise addition
- (ii) $a * b$
 $\nearrow$ dot product (✓)
 $\searrow$ cross product (in MLN)
 vector
 $\downarrow$
 $\vec{a}$

eg. (matrix) $\textcircled{12^{th}}$
 $\Downarrow$

$$a = [a_1 \ a_2 \ a_3 \ a_4]_{1 \times 4}$$

$$b = [b_1 \ b_2 \ b_3 \ b_4]_{1 \times 4}$$

$$a \cdot b = \underbrace{[a_1 \ a_2 \ a_3 \ a_4]}_{(1 \times 4)} \underbrace{[b_1 \ b_2 \ b_3 \ b_4]}_{(1 \times 4)}$$

$\uparrow \quad \uparrow$
 $\textcircled{m \times n} \quad \textcircled{n \times p}$
 Supr

$$\vec{a} \cdot \vec{b} = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix}$$

↓
(1x4)

4x1

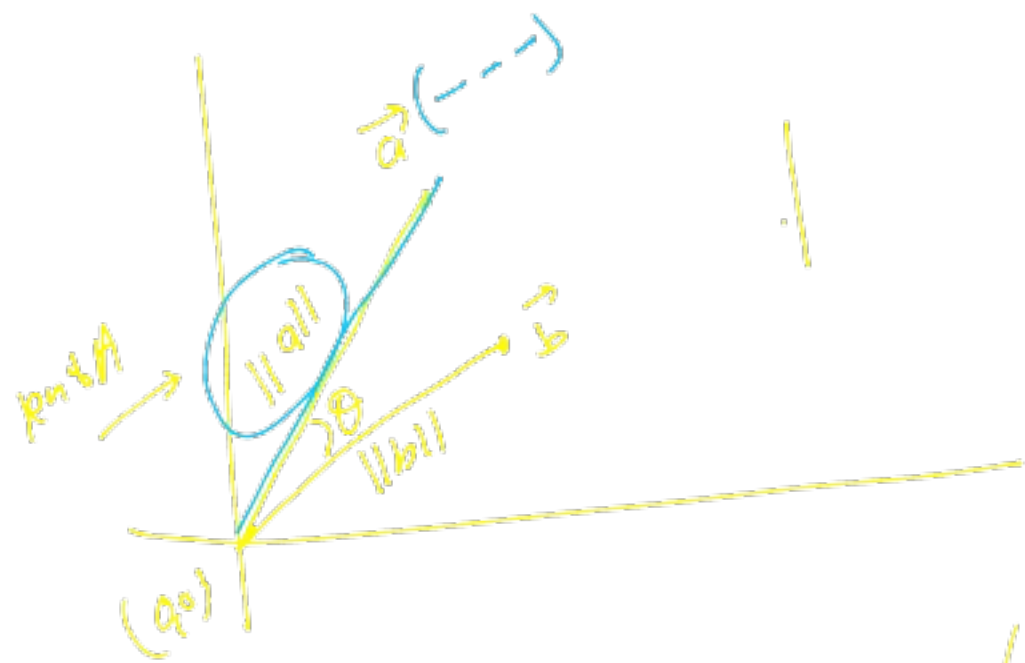
$$\begin{matrix} \vec{a} \\ \vec{b} \end{matrix} \begin{array}{|c|c|c|c|} \hline \textcircled{c_1} & \textcircled{c_2} & \textcircled{c_3} & \textcircled{c_4} \\ \hline 1 & 2 & 3 & 4 \\ \hline 5 & 6 & 7 & 8 \\ \hline \end{array}$$

transpose $\vec{b}$

Row $\rightarrow$ col

$$= a_1 b_1 + a_2 b_2 + a_3 b_3 + a_4 b_4$$

$$\vec{a} \cdot \vec{b} = \underbrace{[1, 2, 3, 4]}_{4d} \cdot \underbrace{\begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}}_{1d} \Rightarrow \dots$$



$$\left\{ \begin{array}{l} \vec{a} = [a_1, a_2] \\ \vec{b} = [b_1, b_2] \\ \vec{a} \cdot \vec{b} = [a_1, a_2] \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \end{array} \right.$$

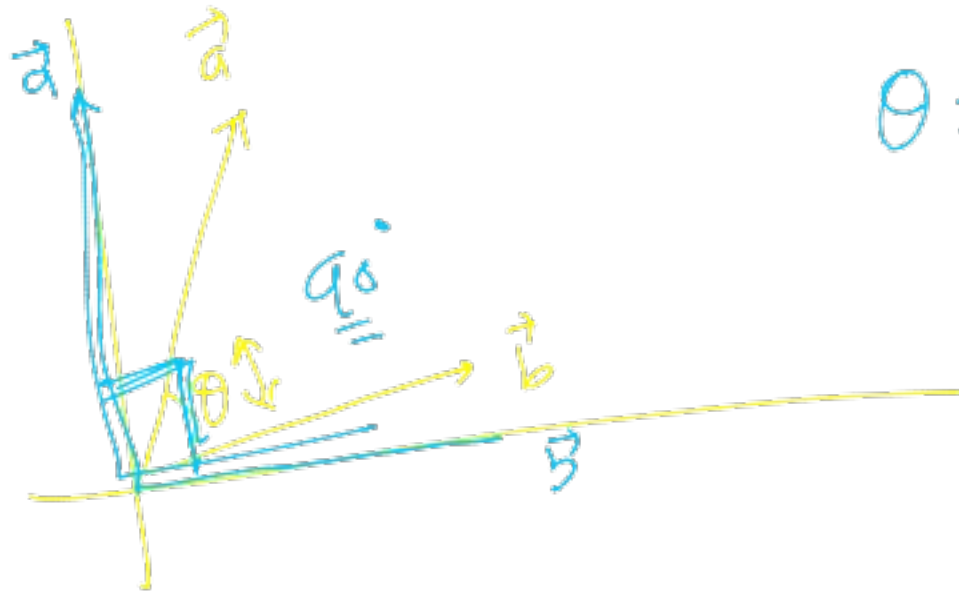
$$\vec{a} = [a_1, a_2, a_3]$$

$$\vec{b} = [b_1, b_2, b_3]$$

$$\vec{a} \cdot \vec{b} = ||\vec{a}|| ||\vec{b}|| \cos \theta \quad (\text{formula})$$

$$a_1 b_1 + a_2 b_2 + a_3 b_3 = ||\vec{a}|| ||\vec{b}|| \cos \theta \rightarrow$$

$$\cos \theta = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{||\vec{a}|| ||\vec{b}||}$$

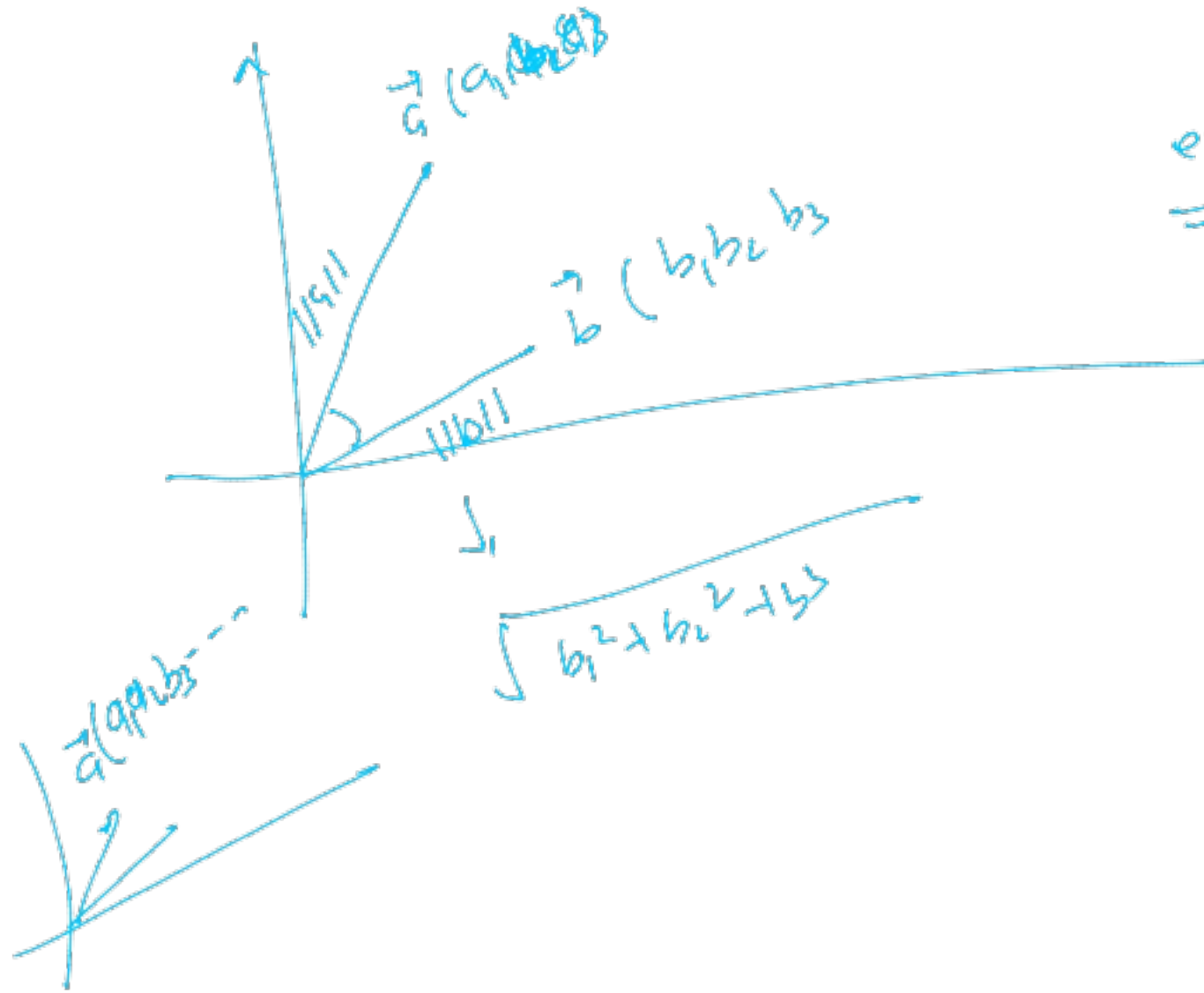


$$\theta = \cos^{-1} \left(\sum_{i=1}^n \right)$$

$$a \cdot b = \|a\| \|b\| \cos q_0$$

$$\boxed{a \cdot b = 0}$$

$$\cos q_0 = 0$$



eg. $P_1 = (0, 4)$
 $P_2 = (4, 3, 5)$

$$\theta = \cos^{-1} \left(\frac{0 + 4 \times 3.5}{\sqrt{16} + \sqrt{16 + 3.5}} \right)$$

$$\theta = 0.5$$

$$\theta = \cos^{-1} \left(\frac{\sum_{i=1}^n a_i b_i}{\|a\| \|b\|} \right)$$

cosine similarity
↓
useful

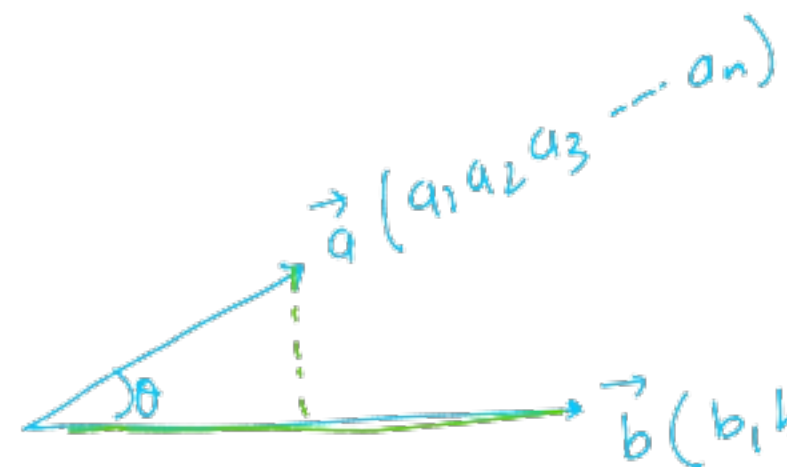
text vectors same
↑

Projection
It,

math

(Q)

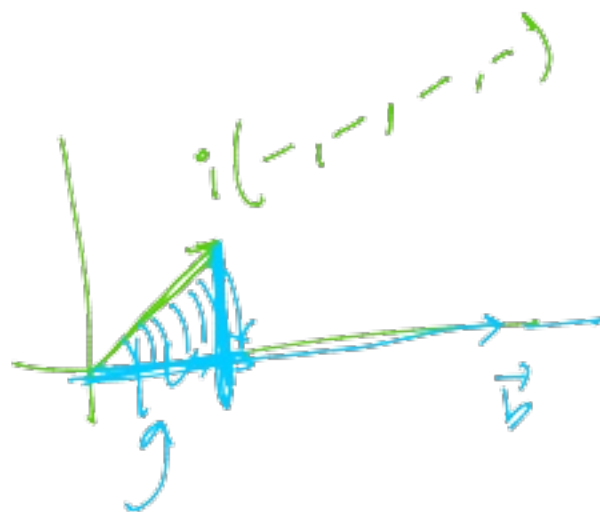
projection of a on
 $b \Rightarrow ?$



Sale
Otr

iphone access

iphone (sales)



~~a~~ b

projection a on b

$$d = \|a\| \cdot \cos \theta$$

— ①

EDA

projection of salary
on year

d_i
 $\downarrow$
 vector
 $\downarrow$
 operation
 $\downarrow$
 car
 $\downarrow$
 project

$\Rightarrow$ Line **
 $\uparrow$

new
 $C_8 \rightarrow [4, 700]$
 =
 BUT Best line

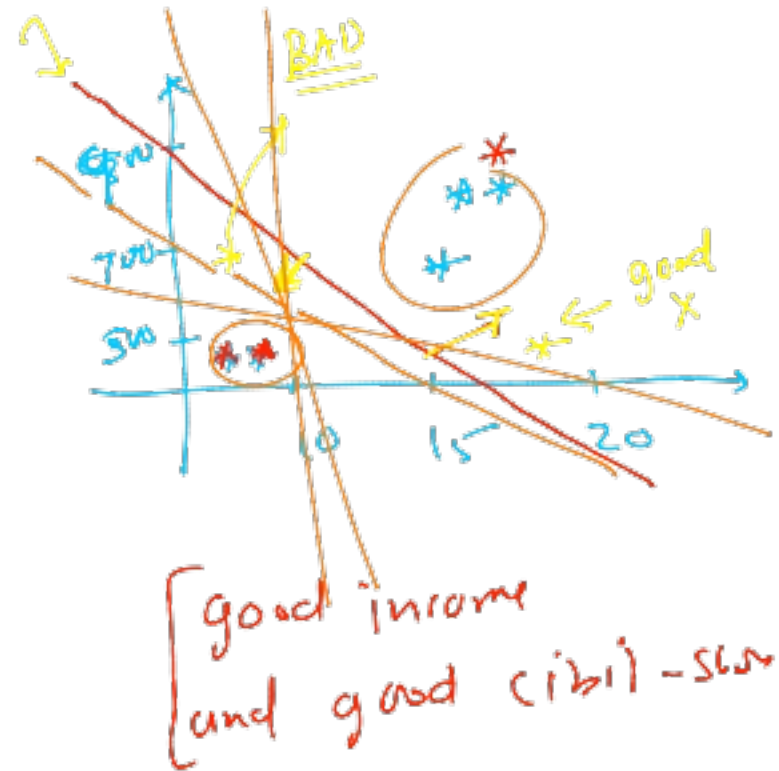
$$\rightarrow$$

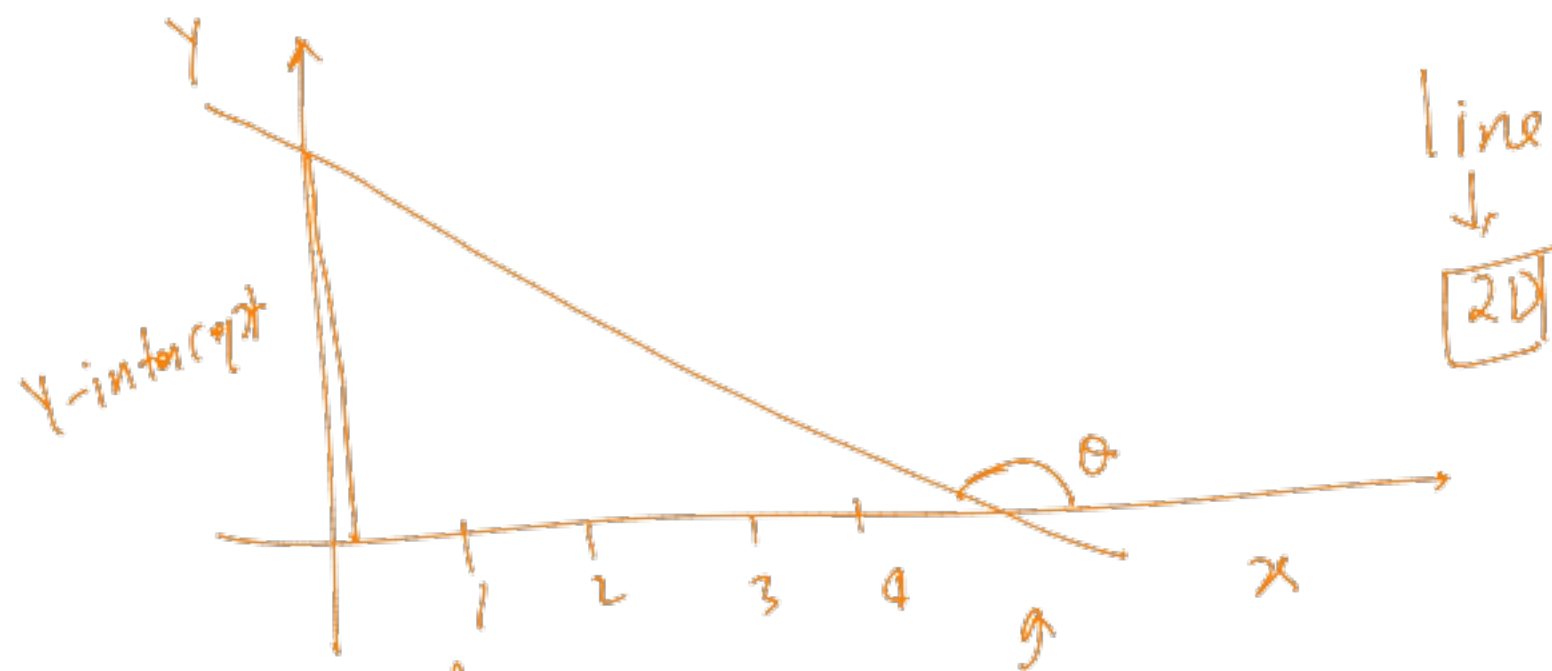
C1	(7	750)
C2	(15	650)
C3	(7	780)
C4	(3	450)
C5	(4	490)

$C_6 \rightarrow [18, 750]$
 $\downarrow$
 good

$C_7 \rightarrow [18, 450]$

Decision surface





at a high level
if you think

slope $[m, c]$

$$y = mx + c$$

↑ slope ↑ y-intercept

$C = 2$
 $m = \text{land } x$
 $m \rightarrow \uparrow$

$m_1 = 3$
 $C_1 = 0.2$

$m_2 = 3.4$
 $C_1 = 0.6$

10 custom 50+
 5 connect

10
 7
 10%

$m_3 = 3.8$
 $C_1 = 0.9$

10 → 9
 90%



cid	x_1	x_2	x_3	x_4
c1	17	750	-	-
c2	16	650	-	-
c3	15	650	-	-
c4	3	450	-	-
c5	4	420	-	-

eqⁿ of line

$$a * x_1 + b * x_2 + c = 0$$

↓

$$\checkmark \quad a * x_1 + b * x_2 + c * x_3 + d = 0$$

↓

3D

Weight

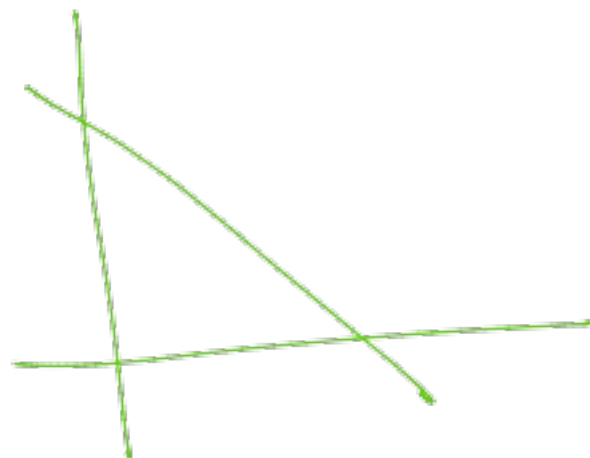
parameter $\rightarrow$?

(slope)

2D $\rightarrow$

line

$$\underset{\substack{\uparrow \\ \text{(code)} \\ \uparrow}}}{w_1} * x_1 + \underset{\uparrow}{w_2} * x_2 + \underset{\uparrow}{w_0} = 0$$



brush

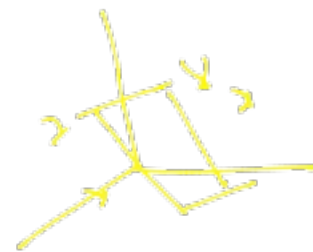
$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 + w_0 = 0$$



3D

$$w_1 * x_1 + w_2 * x_2 + w_3 * x_3 + \dots + w_n * x_n = 0 + w_0$$

4D



n -dimen

2D $\rightarrow$ Line

$\downarrow$
3D $\rightarrow$ plane

$\downarrow$
ND

Hyperspace

data

$$\textcircled{w_1}x_1 + \textcircled{w_2}x_2 + w_3x_3 + \dots + w_nx_n + w_0 = 0$$

$\Downarrow$

$$\begin{bmatrix} w_1 & w_2 & w_3 & w_4 & \dots & w_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_n \end{bmatrix} + w_0 = 0$$

$$\sum_{i=1}^n w_i^T \cdot x_i + w_0 = 0$$



$$\begin{array}{r} [0.15] [90, 760] \\ [0.05] \quad \quad \quad \downarrow \\ \hline \quad \quad \quad \downarrow \\ \text{Score} > 0 \end{array}$$

above the line $\rightarrow$ 4-5
 $\downarrow$
Below $\rightarrow$ NO

general equation of line

↓

→ $ax + by + c = 0$

$$by = -ax - c$$

$$y = \frac{-a}{b}x - \frac{c}{b}$$

m c

eg.

Hand

$$3x + 4y + 1 = 0$$

↓

$$m = -\frac{a}{b}$$

$$m = -\frac{3}{4}$$

$$J = m(x) + C$$

eqⁿ gen

$$ax + by + C = 0$$

$\parallel$
?

x	y
3	✓
4	

$$y = 0.2x + 1$$

m, C

$$\parallel$$

$$\begin{cases} m = 0.2 \\ C = 1 \end{cases}$$



$\text{incont} = [0, 5]$



